Benchmark: **MA.7.GR.1.5** · Scale drawings and scale factor Student code: _____ Calculator: yes

Learning goal: I can solve problems with scale drawings and scale factors. (circle your level after scoring)

4	I can explain how the scale factor changes perimeter versus area and use that in a real-world problem.
3	I can find actual lengths, perimeters, and areas from a scale drawing or scale factor.
2	I can find a scale factor between two figures.
1	I need help to start this benchmark.

1. A rectangle measures 7 units by 10 units. It is enlarged by a scale factor of 2.

What is the area of the enlarged rectangle, in square units?

2. A photo is reduced by a scale factor of $\frac{1}{8}$. A side that measures 72 inches in the original is scaled down.

What is the length of the corresponding side in the reduction, in inches?

Turn over →

Benchmark: **MA.7.GR.1.5** · Scale drawings and scale factor Student code: _____

3. Figure 2 is a scale drawing of Figure 1. A side that measures 2 centimeters in Figure 1 measures 12 centimeters in the corresponding position in Figure 2.

What scale factor was used to produce Figure 2 from Figure 1? Write the scale factor as a whole number or a fraction.

4. A model of a building uses the scale 1 centimeter : 5 meters. On the model, a doorway is 1.5 centimeters long.

How long is the actual doorway, in meters?

5. On a blueprint, the rectangular front wall of a gymnasium measures 4.5 inches long and 5 inches tall. The actual wall is 22.5 feet long.

What is the area of the actual wall, in square feet?

Score: ____ / 5 **My level: 1 2 3 4** **→ copy both to your progress strip**

Answer key — C-GR.1.5 · MA.7.GR.1.5 · Scale drawings and scale factor

Unit 5: Scaling With Proportional Relationships. Regular = on-benchmark item; Rigor = the item that needs the benchmark's reasoning. Mastery on the weekly check = 4 of 5.

Weekly check (5 items)

1. Regular **280 square units**

The enlarged rectangle measures 14 by 20, so its area is 280 square units.

That is $2^2 = 4$ times the original area of 70.

2. Regular **9 inches**

Multiply by the scale factor: $72 \times \frac{1}{8} = 9$ inches.

3. Regular **6**

The scale factor is the constant of proportionality between corresponding lengths: new length ÷ original length.

$12 \div 2 = 6$.

The factor is greater than 1, so Figure 2 is an enlargement.

4. Regular **7.5 meters**

Every centimeter on the model stands for 5 meters, so multiply: $1.5 \times 5 = 7.5$ meters.

5. Rigor **562.5 square feet**

Find the scale factor between the blueprint and the wall: $22.5 \div 4.5 = 5$ feet for each inch.

Actual height: $5 \times 5 = 25$ feet.

Area: $22.5 \times 25 = 562.5$ square feet.

Re-check strip (3 items, C-GR.1.5-R)

1. Regular **14 centimeters**

Multiply the length by the scale factor: $4 \times 3.5 = 14$ centimeters.

2. Regular **B. 64**

Every side is multiplied by 4, and a perimeter is a sum of side lengths, so the perimeter is also multiplied by 4: $16 \times 4 = 64$ centimeters.

Only an area is multiplied by $4^2 = 16$.

3. Regular **4**

The scale factor is the constant of proportionality between corresponding lengths: new length ÷ original length.

$12 \div 3 = 4$.

The factor is greater than 1, so Figure 2 is an enlargement.

Comparison strip (2 items, C-GR.1.5-X) — used on a later week's re-check when this benchmark was *not* retaught

1. Regular **C. 252**

Every side is multiplied by 6, and a perimeter is a sum of side lengths, so the perimeter is also multiplied by 6: $42 \times 6 = 252$ centimeters.

Only an area is multiplied by $6^2 = 36$.

2. Regular **32 centimeters**

Multiply the length by the scale factor: $8 \times 4 = 32$ centimeters.

Reteach plan — C-GR.1.5

MA.7.GR.1.5 · Scale drawings and scale factor

Item that gave the most trouble (from both tally sheets):

What the mistake looks like:

Worked example for Tuesday (8–10 min):

Three practice items (one change at a time):

Decision line: Reteach on _____ Huddle: in person / paper Date: _____ Re-check given on _____

7th Grade Math · Re-check (3 items) C-GR.1.5-R

Benchmark: **MA.7.GR.1.5** · Scale drawings and scale factor Student code: _____ Calculator: yes

1. A drawing is enlarged by a scale factor of 3.5. A side that measures 4 centimeters in the drawing is scaled up.

What is the length of the corresponding side in the enlargement, in centimeters?

2. A rhombus has a perimeter of 16 centimeters. It is enlarged by a scale factor of 4.

What is the perimeter of the enlarged rhombus, in centimeters?

A. 20
B. 64
C. 256
D. 4

3. Figure 2 is a scale drawing of Figure 1. A side that measures 3 centimeters in Figure 1 measures 12 centimeters in the corresponding position in Figure 2.

What scale factor was used to produce Figure 2 from Figure 1? Write the scale factor as a whole number or a fraction.

Score: ____ / 3 → **copy to your progress strip**

7th Grade Math · Re-check (3 items) C-GR.1.5-R

Benchmark: **MA.7.GR.1.5** · Scale drawings and scale factor Student code: _____ Calculator: yes

1. A drawing is enlarged by a scale factor of 3.5. A side that measures 4 centimeters in the drawing is scaled up.

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Score: ____ / 3 → **copy to your progress strip**

7th Grade Math · Comparison items (2)

C-GR.1.5-X

Benchmark: **MA.7.GR.1.5** · Scale drawings and scale factor

Student code: _____ Calculator: yes

1. A trapezoid has a perimeter of 42 centimeters. It is enlarged by a scale factor of 6.

What is the perimeter of the enlarged trapezoid, in centimeters?

A. 1512

B. 48

C. 252

D. 7

2. A drawing is enlarged by a scale factor of 4. A side that measures 8 centimeters in the drawing is scaled up.

What is the length of the corresponding side in the enlargement, in centimeters?

Score: ____ / 2 → **copy to your progress strip**

7th Grade Math · Comparison items (2)

C-GR.1.5-X

Benchmark: **MA.7.GR.1.5** · Scale drawings and scale factor

Student code: _____ Calculator: yes

1. A trapezoid has a perimeter of 42 centimeters. It is enlarged by a scale factor of 6.

What is the perimeter of the enlarged trapezoid, in centimeters?

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C. 252

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Score: ____ / 2 → **copy to your progress strip**